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ACADEMIC VAULT

UNIZIK FRESHER EDITION | 2026

BIO 101: Introduction to Biology - Cells, Genetics and Diversity

The organising ideas of life, from the cell to ecosystems

FACULTY PRIMER

Faculty Primer

Faculty of Bio-Sciences
Botany | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

Study smarter. Ask better questions. Pass with understanding.

Independent study support | Not an official UNIZIK publication

START HERE

Why this pack exists

BIO 101 covers an enormous syllabus, so the danger is memorising isolated facts. This pack organises the course around four unifying ideas - cells, information, energy and evolution - and gives you diagram-and-label practice for the questions that always appear.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Compare prokaryotic and eukaryotic cells and describe the functions of major organelles.
- Explain the cell cycle, mitosis and meiosis and why the distinction matters.
- Apply Mendelian genetics to simple crosses and interpret pedigree information.
- Classify organisms using binomial nomenclature and explain the basis of biological diversity.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

Cells as the unit of life

Every living thing is made of cells, and every cell maintains a boundary, processes energy and stores information.

- Prokaryotes lack a membrane-bound nucleus; eukaryotes have one.
- Nucleus: stores DNA and coordinates gene expression.
- Mitochondria: site of aerobic respiration and ATP production.
- Chloroplasts: photosynthesis in plants and algae.
- Ribosomes: protein synthesis; rough endoplasmic reticulum folds proteins.
- Cell membrane: selectively permeable phospholipid bilayer controlling exchange.

Label diagrams in the exam

Draw large, label with straight lines, and give a function beside each label. Marks are usually split between identification and function.

Cell division and inheritance

Mitosis produces identical cells for growth; meiosis produces gametes with half the chromosome number.

- Mitosis: one division, two diploid daughter cells, genetically identical.
- Meiosis: two divisions, four haploid cells, genetic variation through crossing over and independent assortment.
- Genotype is the genetic makeup; phenotype is the observable trait.
- A dominant allele is expressed when present; a recessive allele is expressed only when homozygous.
- Use a Punnett square for monohybrid crosses and state ratios clearly.
- Pedigrees can indicate whether a trait is dominant, recessive, autosomal or sex-linked.

Energy, ecology and diversity

Life persists by capturing energy and cycling matter through ecosystems.

- Photosynthesis converts light energy into chemical energy; respiration releases it.
- Food chains show energy flow; only about 10 percent passes to the next trophic level.
- Nutrient cycles such as carbon, nitrogen and water connect living and non-living systems.
- Binomial nomenclature: genus capitalised, species lowercase, both italicised.
- Classification hierarchy: domain, kingdom, phylum, class, order, family, genus, species.
- Variation plus selection is the mechanism behind adaptation and speciation.

PRACTICE, NOT PREDICTION

Check your understanding

These questions are written for this pack to test the ideas above. They are not past examination questions and do not predict what your lecturer will set. Answer first, then compare.

1. State one difference between mitosis and meiosis.

Answer: Mitosis produces two genetically identical diploid cells; meiosis produces four genetically varied haploid cells.

2. Why are mitochondria called the powerhouse of the cell?

Answer: They carry out aerobic respiration, producing most of the cell's ATP, the energy currency.

3. Cross Aa with Aa. State the expected genotype ratio.

Answer: AA : Aa : aa = 1 : 2 : 1, giving a 3:1 phenotypic ratio if A is completely dominant.

4. Write the correct binomial format for the human species.

Answer: Homo sapiens - genus capitalised, species lowercase, both italicised.

5. Why is energy lost between trophic levels?

Answer: Organisms use much of their energy for respiration, movement, growth and waste, so only a fraction is stored as biomass available to the next level.

Your 7-step action plan

Tick each step only when you have done it. A tick you cannot defend is worse than an empty box.

- [ ] Draw and label a eukaryotic cell with functions for eight organelles.
- [ ] Make a comparison table of mitosis and meiosis.
- [ ] Solve five monohybrid crosses and one test cross.
- [ ] Learn the classification hierarchy and place five organisms.
- [ ] Draw a food web with at least six organisms and label trophic levels.
- [ ] Define ten biology terms precisely without examples.
- [ ] Answer one past BIO 101 question with diagrams under timed conditions.

Keep going

One pack will not carry a semester. Pair this with your lecture notes, tutorials, and the past questions your department actually releases. If a topic still feels unclear, take a specific question to your lecturer or course rep - "I do not understand the difference between X and Y" gets a better answer than "I do not understand this course".

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YOUR WORKSPACE

Notes, questions and things to ask

Use this page in class or during revision. Write questions you could not answer instead of copying slides.

Three ideas I can now explain without notes:

A question I still cannot answer:

For my lecturer or course rep:

Next revision action and date:

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